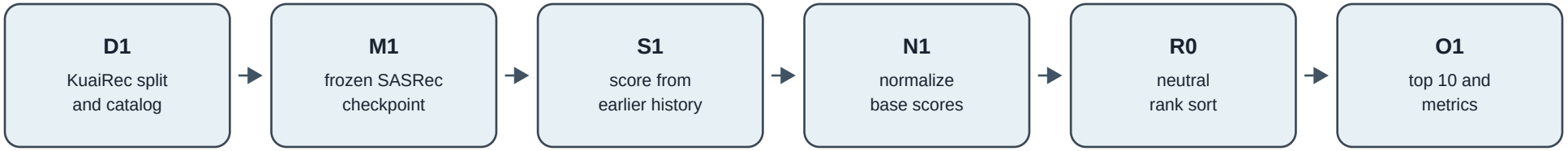
COMP 9500 pipeline and LLM boundary

Code mapping and precise limits: PIPELINE-AND-CODE-WALKTHROUGH.md | FP bec8442 | 27 Oct 2026

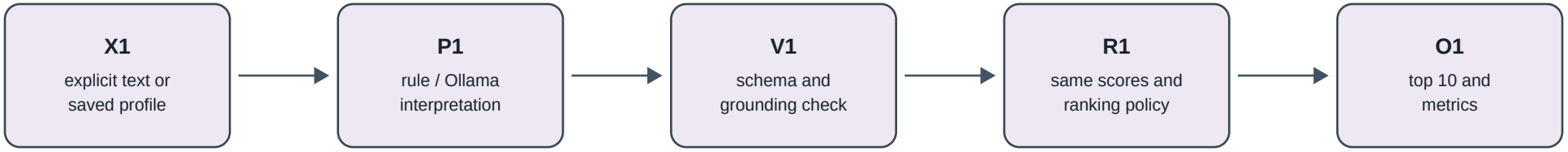
1 Uncontrolled history-only reference (not Condition A after feedback)



2 A/B item feedback (one item; B adds display-only explanations)

3 Independently established human intent: conceptual reference, not a live backend

4 Implemented control language path: rule or local Llama parsing enters at P1



E1 reads O1. U1 displays O1; EX1 can use earlier cached Llama fact selection or a template, without reranking.
A rule-mode feed change proves shared control enforcement, not unique LLM benefit or human satisfaction.

Data, score and comparison paths

Follow the block IDs on page 1; these are implementation paths, not independent experiments.

D1 KuaiRec input and split

feedctrl/data.py:91-194 builds the processed catalog, user histories and split metadata. Training, request-history and final-test windows are separate. Synthetic E1 requests use earlier history; the held-out outcomes do not become real user intent. Check the saved manifest before reproducing an evaluation.

M1 and S1 Frozen model and base scores

feedctrl/model.py:26-88 defines an independently written SASRec variant; model.py:203-237 scores catalog items and loads a saved checkpoint. DemoState.rank in feedctrl/api.py:140-149 scores the fixed sorted candidate catalog from earlier history. The saved seed-42 checkpoint was used for the worked example. Diagnostic scoring is a separately labelled fallback, not the trained model.

N1 and R0 Untouched history-only reference

controls.normalize_scores at feedctrl/controls.py:297-305 normalizes base scores. Empty command and profile produce a neutral sort through rerank at lines 326-339. R0 is the history-only calculation; current Condition A is a different comparator once item feedback is applied.

A1 Manual UI feedback proxy

The A/B path adds a like to one item or removes that preference for dislike (api.py:150-157, /api/feedback at 295-305). evaluation.item_feedback_ranking at 75-84 is the offline proxy. KuaiRec does not provide observed thumbs labels; this is not a historical ground-truth thumb.

H0 Human-intent reference is conceptual

A person would independently record an intended operation and category, then that structured command could enter V1 and the same R1 policy. No separate human-parser backend or reserved human-reference ranking run is claimed. The completed student 100-response worksheet, 99/100 comparison and separate 73/100 versus 18/100 exposed-practice parser check are descriptive, not a fresh held-out human ranking evaluation.

From language to a changed feed

Only P1 is the candidate language-understanding advantage; the recommender and enforcement are shared.

X1 and P1 Text, profile, rule or Ollama

api.py:243-293 accepts explicit control text and editable profile text. controls.parse_control at 243-254 and parse_profile at 257-294 select a rule parser or local Ollama. The current live app is grounded to one explicit category_N ID, not arbitrary named topics. A parser response is a proposed structured action, not a feed change by itself.

V1 Validate before state changes

controls._validate_control at 104-119 checks schema and explicit allowed category grounding. Invalid or ambiguous language fails or asks for clarification. An Ollama error returns HTTP 502 and preserves prior state; there is no silent rule-mode success. A valid structured output can still disagree with intended action.

R1 and O1 Shared policy, rank and response

controls.apply_to_profile at 308-323 gives an explicit same-category command precedence. rerank at 326-339 adds category boost weights and hard-excludes muted-category items before sorting. The top 10, scores, preference counts and notices are returned through feedctrl/api.py. The rule and LLM paths use the same scoring and policy after validation.

A-E conditions and explanation boundary

A: one-item feedback. B: A plus display-only explanations, so order stays equal. C: parsed free-text category control. D: saved explicit profile preference. E: profile plus explicit control; the later same-category command wins rather than double-counting. EX1 in feedctrl/explanations.py:20-86 selects constrained saved facts or a labelled template; an earlier cached Llama selection may be present, but display does not rerank and human response was not measured.

One executed rule-mode example, not a new science result

With saved processed data, user 14 and seed-42 model: neutral top three 8958, 7383, 1116. Like item 1116 makes it first in A and B. Explicit category_1 boost yields nine of ten target items in matched C/D/E; muting category_1 in E removes them. The receipt and reproducer are in the companion packet. No fresh LLM call or browser action is implied.

Evaluation and what the result means

Use the numbers with their protocol and denominator; avoid treating an exposure metric as human satisfaction.

E1 Metric calculation

evaluation.ranking_metrics at 51-72 evaluates fixed top K=10: TCP is target-category item count divided by ten for positive requests; TCER is the same exposure fraction for mutes. Fill reports how many slots remain occupied. HR@1 and NDCG@10 compare against historical next-item relevance. User-level paired summaries average three fixed seeds; request-seed rows are not independent users.

Frozen Experiment 1

Canonical rule and Llama rankings are identical, so E1 feed changes show shared category-policy enforcement, not an LLM-specific gain. TCP rose 11.6% to 79.2% and TCER fell 6.1% to zero under their assigned requests; boost NDCG@10 fell 45.7%. Broader synthetic parser accuracy was Llama 79/100 versus rules 81/100, below the proposal's 85% target.

Revised AI-reference Experiment 2 and separate student check

The completed fresh AI-reference E2 reports whole-command agreement Llama 76/100 versus rules 11/100 and induced success 20.145% versus item-feedback 3.815%; NDCG was 0.02580 versus 0.03184 and conditional success was zero. References are model-generated. The separately dated student-label exposed-practice parser result is 73/100 versus 18/100; it is not the same fresh evaluation.

Validity boundary and professor's decision

Offline TCP/TCER can verify exposure under an assigned category command. They do not establish that a person wanted that command, found the result useful, or remained satisfied. The original archived human-reference route was not executed; the revised AI-reference study is complete. A new independently specified human-intent protocol is a proposed next study only if the supervisor requests it.